

Distribution of Singular Data Generated by Compact Forced Navier–Stokes Blowup

Zhuoni Chi
znchi@zju.edu.cn

Shaozhen Cao
shaozhencao@zju.edu.cn

9 September 2026

Abstract

Starting from the compact, smoothly forced Navier–Stokes blowup solution constructed by OpenAI, we study the distribution of the singular data it generates. For fixed viscosity and time horizon on the three-dimensional torus, smooth forces producing classical breakdown from rest by that time are dense in the inherited time-integrated spatial H^s topology if and only if $s < 1/2$ (the norms are defined in Section 1). The positive result follows from an exact insertion into every regular reference trajectory. A localized vector potential removes the background around a concentrated singular packet, so all nonlinear cross terms vanish and the modified force remains smooth through the singular time. We give the full support construction, derivative estimates, fractional Sobolev scaling and classical-lifespan argument. The inserted trajectories converge strongly in the energy and dissipation norm. A separate critical-force bootstrap, followed by an H^1 estimate and high-Sobolev continuation, supplies the regular open set needed for non-density at and above $s = 1/2$. Further results describe extended-data projections, every fixed smooth initial-velocity slice, mixed force norms, infinite-dimensional variations and interior no-slip realizations. The construction varies the force; it does not classify singular initial velocities for a single prescribed force.

Keywords: forced Navier–Stokes equations; singular data; localized insertion; critical Sobolev topology; energy convergence.

1 Introduction and main result

The initial datum of a forced evolution equation is only one part of its input. The external force is another. Accordingly, the distribution of singular inputs can mean the distribution of initial velocities at fixed force, the distribution of forces at fixed initial velocity, or the distribution of pairs. A singular solution starting from rest has just one initial velocity, even if its force admits a large family of variations.

OpenAI’s *Finite Time Blowup for Navier–Stokes* [1, Theorem 1.1] supplies, for every positive viscosity, a compact whole-space solution with zero initial velocity, globally smooth compactly supported forcing, bounded kinetic energy and unbounded speed at time one. We take this theorem and its associated Lean formalization as established results. The compact packet is restated in Theorem 3.1. The present article proves distribution properties of the data obtained from that packet.

Hofmanová, Zhu and Zhu [3, Theorem 1.4, Corollary 4.6 and Theorem 5.1] provide a close antecedent: forces producing nonunique Leray–Hopf solutions are dense in $L^1(0, 1; L^2(\mathbb{R}^3))$ from rest, and every L^2_σ initial velocity admits a force producing nonuniqueness. This motivates our emphasis on force topology and on fixing the initial velocity while varying the force. Here the target property is classical breakdown under globally smooth forcing, and the construction removes a regular background locally before inserting the compact packet. The weak-solution

nonuniqueness theorem does not supply that packet or the sharp Sobolev thresholds proved here; our conclusions do not assert nonuniqueness of Leray–Hopf continuations.

The analytic prerequisites are elementary calculus, Lebesgue and Bochner integration, Fourier inversion and Plancherel, Hölder’s inequality, and the contraction principle. The critical embeddings are proved in Appendix A; the multiplication estimates, local evolution and continuation arguments used here are proved in the text. The other references provide historical and conceptual context, rather than additional PDE existence inputs.

Fix $\nu > 0$ and $T > 0$. The principal domain is the unit torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, with volume one.

For $s \in \mathbb{R}$ and a periodic scalar or vector distribution z , write $\widehat{z}(k) = \int_{\mathbb{T}^3} z(x) e^{-2\pi i k \cdot x} dx$, with Fourier coefficients understood distributionally when necessary. The inhomogeneous space $H^s(\mathbb{T}^3)$ consists of distributions with finite norm

$$\|z\|_{H^s}^2 = \sum_{k \in \mathbb{Z}^3} (1 + 4\pi^2 |k|^2)^s |\widehat{z}(k)|^2.$$

The homogeneous space $\dot{H}^s(\mathbb{T}^3)$ consists of mean-zero distributions with finite norm

$$\|z\|_{\dot{H}^s}^2 = \sum_{k \neq 0} (2\pi |k|)^{2s} |\widehat{z}(k)|^2.$$

The homogeneous and inhomogeneous norms are equivalent on mean-zero fields at each fixed order. Vector and tensor norms are the square root of the sum of the squared component norms. Unless a domain is shown, a spatial norm in the periodic argument is taken on $\mathbb{T}^3$.

For a strongly measurable map $z : I \rightarrow X$ into a normed spatial space, use

$$\|z\|_{L^q(I; X)} = \left(\int_I \|z(t)\|_X^q dt \right)^{1/q} \quad (1 \leq q < \infty), \quad \|z\|_{L^\infty(I; X)} = \operatorname{ess\,sup}_{t \in I} \|z(t)\|_X. \quad (1)$$

Thus $L_t^1 H_x^s$ for forces means $L^1(0, \infty; H^s(\mathbb{T}^3))$, with norm $\int_0^\infty \|f(t)\|_{H^s(\mathbb{T}^3)} dt$; H^s measures spatial regularity only. Unless an interval is displayed, mixed force norms use $(0, \infty)$ and presingular velocity norms use $(0, T)$. The same convention applies to $X = L^p$ and $X = \dot{H}^s$, and to the explicitly named domains below.

We consider

$$\partial_t u + (u \cdot \nabla) u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad u(0) = a. \quad (2)$$

Both u and p are periodic; pressure is normalized by $\int_{\mathbb{T}^3} p = 0$. The smooth input spaces are

$$\mathcal{X} = C_{\operatorname{div}}^\infty(\mathbb{T}^3; \mathbb{R}^3), \quad \mathcal{F} = C_c^\infty(\mathbb{T}^3 \times (0, \infty); \mathbb{R}^3). \quad (3)$$

The support restriction in $\mathcal{F}$ concerns positive time. In particular, each force is defined and smooth through any possible terminal time of its velocity.

Let $T_{\max}^\nu(a, f)$ denote the maximal classical lifespan for the pressure-normalized problem. Its existence and uniqueness framework is recalled in Proposition 2.2. Define

$$\mathcal{B}_{\nu, a, T} = \{f \in \mathcal{F} : T_{\max}^\nu(a, f) \leq T\}, \quad \mathcal{B}_{\nu, T}^0 = \mathcal{B}_{\nu, 0, T}. \quad (4)$$

Theorem 1.1 (Sobolev density threshold). *Equip $\mathcal{F}$ with its relative $L^1(0, \infty; H^s(\mathbb{T}^3))$ norm topology.*

- (i) *For every fixed $a \in \mathcal{X}$, the set $\mathcal{B}_{\nu, a, T}$ is dense if $s < 1/2$.*
- (ii) *On the zero-initial-velocity slice,*

$$\mathcal{B}_{\nu, T}^0 \text{ is dense in } \mathcal{F} \iff s < \frac{1}{2}.$$

Here density is relative to smooth forces in a specified norm. It is not density in the usual test-function topology, which imposes control of all derivatives on a common support. The approximating forces below concentrate in shrinking regions and have increasing peak amplitudes.

The construction is local in space and late in time. Given a reference solution regular through T , we preserve its initial velocity and its history until shortly before T , then insert a packet that becomes singular exactly at T . The resulting force is smooth on the entire positive time axis. The velocity perturbation tends to zero in

$$\|z\|_{E_T} := \|z\|_{L^\infty(0,T;L^2)} + \|\nabla z\|_{L^2(0,T;L^2)}. \quad (5)$$

This is equivalent, on a fixed finite interval, to the sum norm of $L_t^\infty L_x^2 \cap L_t^2 H_x^1$. It requires neither a classical value nor a continuation at T .

The proof separates three issues. First, parabolic concentration makes the packet small in subcritical force norms. Second, a divergence-free background correction makes the nonlinear insertion exact while keeping its force smooth at the singular endpoint. Third, a separate critical regularity estimate proves the existence of an open regular neighborhood. A scaling calculation alone would not establish non-density.

Section 2 fixes the analytic conventions and the continuation criterion. Section 3 derives the packet estimates, including the fractional localization bound. Section 4 proves the insertion theorem and its interior-boundary extension. Section 5 treats the density and projection quantifiers. Section 6 establishes the critical obstruction and completes Theorem 1.1. Section 7 gives additional families.

2 Analytic conventions and classical continuation

2.1 Sobolev norms, projection and pressure

Euclidean and interior-domain norms. For a scalar distribution Z on $\mathbb{R}^3$, our Fourier convention is $\widehat{Z}(\xi) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix \cdot \xi} Z(x) dx$, initially on Schwartz functions and then by duality. For an open domain $\Omega \subset \mathbb{R}^3$ and every $s \in \mathbb{R}$ set

$$\|Z\|_{H^s(\mathbb{R}^3)}^2 = \int_{\mathbb{R}^3} (1 + |\xi|^2)^s |\widehat{Z}(\xi)|^2 d\xi, \quad \|z\|_{H^s(\Omega)} = \inf_{Z|_{\Omega}=z} \|Z\|_{H^s(\mathbb{R}^3)}. \quad (6)$$

Here $H^s(\mathbb{R}^3)$ consists of tempered distributions with finite displayed norm, and $H^s(\Omega)$ is its space of distributional restrictions with the quotient norm. In particular, negative orders use this restriction definition, not a separately chosen dual-space convention. At $s = 0$ this is the usual $L^2(\Omega)$ norm. The auxiliary Euclidean homogeneous norm is $\|Z\|_{\dot{H}^s(\mathbb{R}^3)}^2 = \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{Z}(\xi)|^2 d\xi$ whenever finite; below it is used only for nonnegative orders on compact smooth profiles in the scaling argument. It does not replace the inhomogeneous norm in the force topology.

For every fixed compact $K \Subset \Omega$ and $s \in \mathbb{R}$, smooth fields supported in K obey

$$\|z\|_{H^s(\Omega)} \leq \|E_0 z\|_{H^s(\mathbb{R}^3)} \leq C_{s,K,\Omega} \|z\|_{H^s(\Omega)}, \quad \text{supp } z \subset K, \quad (7)$$

where $E_0 z$ is extension by zero. To prove the second inequality, fix $\chi \in C_c^\infty(\Omega)$ equal to one near K . For every extension Z of z , $\chi Z = E_0 z$. Multiplication by this fixed χ is bounded on $H^s(\mathbb{R}^3)$ for every real s : the Fourier convolution formula and

$$\frac{(1 + |\xi|^2)^{s/2}}{(1 + |\eta|^2)^{s/2}} \leq C_s (1 + |\xi - \eta|^2)^{|s|/2}$$

reduce the estimate to Young's $L^1 * L^2 \rightarrow L^2$ inequality, since $\widehat{\chi}$ decays faster than every power. Taking the infimum over Z proves the bound; the first inequality follows from the definition.

The constant depends on the fixed cutoff and s , not on a smaller support or a shrinking scale ε . Applying the pointwise bounds in time gives the same comparison for all the mixed norms in (1), provided K is fixed for all times. No bounded zero-extension operator on arbitrary $H^s(\Omega)$ is asserted. All vector and tensor norms in these definitions are the square root of the sum of the squared component norms.

The Leray projection $\mathbb{P}$ is the Fourier multiplier

$$\widehat{\mathbb{P}z}(k) = \left(I - \frac{k \otimes k}{|k|^2}\right) \widehat{z}(k) \quad (k \neq 0), \quad \widehat{\mathbb{P}z}(0) = \widehat{z}(0).$$

It is bounded on every H^s , commutes with spatial derivatives and the heat semigroup, and eliminates periodic gradients. Equation (2) becomes

$$\partial_t u - \nu \Delta u = -\mathbb{P} \nabla \cdot (u \otimes u) + \mathbb{P} f. \quad (8)$$

The pressure is then recovered uniquely by

$$\Delta p = \nabla \cdot f - \sum_{i,j} \partial_i \partial_j (u_i u_j), \quad \int_{\mathbb{T}^3} p = 0.$$

The right side has zero mean. Periodicity of the pressure is part of the solution class; a nonperiodic affine pressure is not an allowed alternative gauge.

Lemma 2.1 (Sobolev estimates used below). *For each integer $k \geq 3$ there are constants depending only on k and the torus such that*

$$\|zw\|_{H^k} \leq C_k \|z\|_{H^k} \|w\|_{H^k}, \quad (9)$$

$$\|u \otimes u\|_{H^k} \leq C_k \|u\|_{H^2} \|u\|_{H^k}. \quad (10)$$

For mean-zero v ,

$$\|v\|_3 \leq C \|v\|_{\dot{H}^{1/2}}, \quad \|\nabla v\|_3 + \|(-\Delta)^{1/2} v\|_3 \leq C \|v\|_{\dot{H}^{3/2}}. \quad (11)$$

We also use $H^1 \hookrightarrow L^6$ and $H^2 \hookrightarrow L^\infty$.

Proof. Write $\langle \ell \rangle = (1 + 4\pi^2 |\ell|^2)^{1/2}$. The triangle inequality in $\mathbb{R}^4$ and the elementary power inequality give

$$\langle \ell \rangle^r \leq C_r (\langle j \rangle^r + \langle \ell - j \rangle^r) \quad (r \geq 0).$$

Using the latter inequality with $r = k$ in the Fourier convolution of zw yields

$$\|zw\|_{H^k} \leq C_k (\|\widehat{z}\|_{\ell^1} \|w\|_{H^k} + \|\widehat{w}\|_{\ell^1} \|z\|_{H^k}).$$

Here the sequence convolution estimate follows directly by writing an $\ell^1 * \ell^2$ convolution as a sum of translates and applying the triangle inequality in ℓ^2 . Cauchy–Schwarz gives

$$\|\widehat{z}\|_{\ell^1} \leq \left(\sum_{\ell \in \mathbb{Z}^3} \langle \ell \rangle^{-4} \right)^{1/2} \|z\|_{H^2} \leq C \|z\|_{H^2}.$$

The sum converges: a dyadic shell of radius 2^j contains at most $C2^{3j}$ lattice points and its weighted contribution is at most $C2^{-j}$. This proves (9) and (10), initially for Fourier polynomials and then by their Fourier-norm approximation. The same absolute Fourier convergence proves $\|z\|_\infty \leq C \|z\|_{H^2}$.

Lemma A.1 in Appendix A proves the critical embeddings on both domains. Apply its order-1/2 inequality to v , each component of ∇v , and $(-\Delta)^{1/2} v$ to obtain (11). Its order-one inequality applied to $z - \widehat{z}(0)$, together with $|\widehat{z}(0)| \leq \|z\|_2$, gives the inhomogeneous $H^1 \hookrightarrow L^6$ assertion. Thus no integer Gagliardo–Nirenberg estimate is needed here. $\square$

Proposition 2.2 (Local solutions, uniqueness and continuation). *Let $a \in \mathcal{X}$ and let f be periodic and smooth on every compact interval in $[0, \infty)$. There is a unique maximal smooth solution of (2), with normalized periodic pressure, on $[0, T_{\max})$. If $S < \infty$ and*

$$\int_0^S \|u(t)\|_{H^2}^2 dt < \infty \quad (12)$$

on a classical solution defined for $0 \leq t < S$, then that solution extends smoothly beyond S .

Proof. One common H^3 interval. Let $J = (1 - \Delta)^{1/2}$, defined by the Fourier weights above. For $r \geq 0$,

$$\|e^{\nu t \Delta} \mathbb{P} \nabla \cdot A\|_{H^r} \leq C(\nu t)^{-1/2} \|A\|_{H^r}, \quad t > 0,$$

because the multiplier has norm at most $\sup_{\xi \in \mathbb{R}^3} |\xi| e^{-\nu t |\xi|^2} \leq C(\nu t)^{-1/2}$. The heat semigroup is a contraction and is strongly continuous on every H^r , as follows by dominated convergence in its Fourier series. Consider the mild equation

$$\begin{aligned} u(t) &= e^{\nu t \Delta} a - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (u \otimes u)(s) ds \\ &\quad + \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} f(s) ds. \end{aligned} \quad (13)$$

It maps the space of real, divergence-free fields in $C([0, \tau]; H^3)$ to itself. The integral is continuous in time: split off an interval of length h next to the upper limit, whose norm is $O(h^{1/2})$, and use semigroup strong continuity on the remaining interval. Lemma 2.1 bounds the nonlinear integral by $C\nu^{-1/2}\tau^{1/2} \|u\|_{CH^3}^2$, and the difference of two such integrals by

$$C\nu^{-1/2}\tau^{1/2} (\|u\|_{CH^3} + \|v\|_{CH^3}) \|u - v\|_{CH^3}.$$

Take $R = 2(\|a\|_{H^3} + 1)$ and choose $\tau > 0$ so that $\int_0^\tau \|f(s)\|_{H^3} ds \leq 1$ and $2C\nu^{-1/2}\tau^{1/2}R \leq 1/2$. The map preserves the closed ball of radius R and is a contraction there. Its iterates are Cauchy by a geometric estimate in the complete space $C([0, \tau]; H^3)$; their limit is the unique fixed point in that ball.

Orthogonal truncations and uniform higher-order bounds. For an integer $N \geq 1$ define

$$\widehat{Q_N z}(\ell) = \mathbf{1}_{\{|\ell| \leq N\}} \widehat{z}(\ell).$$

This is a self-adjoint orthogonal projection, has norm at most one on every H^r , and commutes with J , $\mathbb{P}$, spatial derivatives and the heat semigroup. Its symmetric frequency set preserves real fields. On exactly the same CH^3 ball and interval, the projected mild map gives $u_N = Q_N u_N$ solving

$$\partial_t u_N - \nu \Delta u_N = -Q_N \mathbb{P} \nabla \cdot (u_N \otimes u_N) + Q_N \mathbb{P} f, \quad u_N(0) = Q_N a, \quad \|u_N\|_{CH^3} \leq R.$$

Differentiating the finitely many mild coefficient equations first gives C^1 coefficients solving a polynomial ordinary differential equation; repeated differentiation, since the forcing is smooth, gives C^∞ coefficients. Thus u_N is smooth and the following energy identities are justified directly. For an integer $k \geq 3$ put $X_{k,N} = \|u_N\|_{H^k}$ and $D_{k,N} = \|\nabla u_N\|_{H^k}$. Apply J^k , pair with $J^k u_N$, and move both projections to u_N using self-adjointness. Integrating the nonlinear divergence by parts and using (10) gives

$$\frac{1}{2} (X_{k,N}^2)' + \nu D_{k,N}^2 \leq C_k \|u_N\|_{H^2} X_{k,N} D_{k,N} + \|f\|_{H^k} X_{k,N}.$$

The dissipation is the displayed gradient norm; no Poincaré estimate on the possibly nonzero mean is used. Young's inequality and division regularized by $(X_{k,N}^2 + \zeta^2)^{1/2}$ imply, after integration and $\zeta \downarrow 0$,

$$X_{k,N}(t) \leq e^{C_{k,\nu} R^2 \tau} \left(\|a\|_{H^k} + \int_0^\tau \|f(s)\|_{H^k} ds \right) =: M_k.$$

Every M_k is independent of N . In particular, higher regularity does not require shrinking the original H^3 interval.

Strong convergence, without a compact embedding. Subtract the mild equation for u from the projected equation. With the same contraction constant $q \leq 1/2$, the difference estimate is

$$\begin{aligned} \|u_N - u\|_{CH^3} &\leq q \|u_N - u\|_{CH^3} + \|(I - Q_N)a\|_{H^3} + \int_0^\tau \|(I - Q_N)\mathbb{P}f(s)\|_{H^3} \, ds \\ &\quad + C\nu^{-1/2}\tau^{1/2} \|(I - Q_N)(u \otimes u)\|_{CH^3}. \end{aligned}$$

The last three terms tend to zero. For the last one, $u \otimes u$ is continuous into H^3 , so its image of $[0, \tau]$ is compact. Strong convergence $Q_N \rightarrow I$ is uniform on that image: cover it by finitely many balls of radius ε , use convergence at their centers, and use $\|I - Q_N\| \leq 1$. The forcing term follows either in the same way or by dominated convergence. Hence $u_N \rightarrow u$ in CH^3 .

At each time, convergence of the Fourier coefficients and Fatou's lemma transfer $\|u_N(t)\|_{H^k} \leq M_k$ to $\|u(t)\|_{H^k} \leq M_k$. For $3 < m < k$, Hölder's inequality in the Fourier sum gives the explicit interpolation bound

$$\|w\|_{H^m} \leq \|w\|_{H^3}^{(k-m)/(k-3)} \|w\|_{H^k}^{(m-3)/(k-3)}.$$

Indeed each squared weighted Fourier term is the product of the order-three and order- k terms raised to those two exponents; summing and applying Hölder proves the formula. Apply it to $w = u_N - u$, whose H^k norm is at most $2M_k$, and choose an integer $k > m$. We obtain convergence in $C([0, \tau]; H^m)$ for every fixed integer $m \geq 0$ (orders at most three follow directly). Thus u has continuous spatial regularity of every order on the same interval.

Time regularity and pressure. The right sides of the truncated evolution equations converge in CH^m for every m , using the just-proved convergence in CH^{m+3} , the product estimate and uniform projection convergence on continuous images. Passing to their time integrals gives

$$u \in C^1([0, \tau]; H^m), \quad \partial_t u = \nu \Delta u - \mathbb{P} \nabla \cdot (u \otimes u) + \mathbb{P} f.$$

Time derivatives at zero are right derivatives. Repeatedly differentiate this identity: the derivative of the product is a finite sum of products of previously obtained derivatives. Induction, with arbitrarily high spatial order available at each step, gives $u \in C^j([0, \tau]; H^m)$ for all j, m . The pressure formula preceding the lemma has, at nonzero frequencies, order -1 on f and order zero on $u \otimes u$; its zero coefficient is set to zero. It therefore gives the same time and spatial regularity for p . Finally, for $m > r + 3/2$, Cauchy–Schwarz makes the Fourier series of all spatial derivatives of order at most r absolutely and uniformly convergent, since $\sum_\ell \langle \ell \rangle^{-2(m-r)} < \infty$. Applied also to time derivatives, this proves joint classical smoothness on every such closed interval.

Uniqueness and the maximal interval. If u_1, u_2 are two smooth solutions with the same data and $z = u_1 - u_2$, then

$$\partial_t z - \nu \Delta z + (u_1 \cdot \nabla)z + (z \cdot \nabla)u_2 + \nabla(p_1 - p_2) = 0.$$

Testing against z and using incompressibility and periodicity gives

$$\frac{1}{2}(\|z\|_2^2)' + \nu \|\nabla z\|_2^2 \leq \|\nabla u_2\|_\infty \|z\|_2^2.$$

Grönwall yields $z = 0$ on every closed common interval, and the pressure normalization identifies the pressures. Restarting the construction at any smooth time and joining these unique solutions defines the maximal interval $[0, T_{\max})$.

Continuation. The same energy calculation on a smooth solution, now writing $X_k = \|u\|_{H^k}$ and $D_k = \|\nabla u\|_{H^k}$, gives

$$\frac{1}{2}(X_k^2)' + \nu D_k^2 \leq C_k \|u\|_{H^2} X_k D_k + \|f\|_{H^k} X_k.$$

Young's inequality and regularized division therefore give the integrated form of

$$X'_k \leq C_{k,\nu} \|u\|_{H^2}^2 X_k + \|f\|_{H^k}. \quad (14)$$

Under (12), Grönwall bounds X_k uniformly for $t < S$ at every fixed k , since f is smooth on $[0, S+1]$. The H^3 construction gives a positive lower bound for the restart interval using only a bound for $\|u(t_0)\|_{H^3}$ and $\sup_{[0, S+1]} \|f\|_{H^3}$. Choose $t_0 < S$ closer to S than that interval length (and shorten the length to at most one). The restarted solution extends beyond S , has all higher regularity by the common interval argument, and equals the original solution on their overlap by uniqueness. This proves continuation without assuming an endpoint value in H^3 . The criterion proved here is precisely (12); the unbounded speed of the inserted solutions below follows separately from the packet theorem. $\square$

3 The compact packet and its scaling

Theorem 3.1 (Compact forced blowup packet, OpenAI). *At the fixed viscosity $\nu > 0$, there exist smooth fields U, P on $\mathbb{R}^3 \times [0, 1)$, a force $F \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$, and a compact set K such that*

$$\partial_t U + (U \cdot \nabla)U - \nu \Delta U + \nabla P = F, \quad \nabla \cdot U = 0, \quad U(0) = 0, \quad (15)$$

$$\text{supp } U(t) \cup \text{supp } P(t) \subset K \quad (0 \leq t < 1),$$

$$\sup_{0 \leq t < 1} \|U(t)\|_2 < \infty, \quad \limsup_{t \uparrow 1} \|U(t)\|_\infty = \infty. \quad (16)$$

This is [1, Theorem 1.1]. Its associated formalization [2] supplies the zero-data witness and compact-support properties. We use precisely the displayed packet properties to derive the estimates and distribution results below.

Lemma 3.2 (Energy, dissipation and initial vanishing). *The packet satisfies*

$$M := \sup_{0 \leq t < 1} \|U(t)\|_2 < \infty, \quad D := \|\nabla U\|_{L^2((0,1) \times \mathbb{R}^3)} < \infty.$$

Both U and P vanish on an initial time interval and therefore extend smoothly by zero to negative times.

Proof. On each closed interval $[0, b]$ with $b < 1$, compact spatial support justifies the energy identity

$$\frac{1}{2}(\|U(t)\|_2^2)' + \nu \|\nabla U(t)\|_2^2 = \langle F(t), U(t) \rangle.$$

Set $N(t) = \int_0^t \|F(s)\|_2 \, ds$. Dropping dissipation gives $\frac{1}{2}(\|U\|_2^2)' \leq \|F\|_2 \|U\|_2$. For $\zeta > 0$, the derivative of $(\|U\|_2^2 + \zeta^2)^{1/2}$ is at most $\|F\|_2$. Integrating from zero and letting $\zeta \downarrow 0$ proves $\|U(t)\|_2 \leq N(t)$.

Substituting this bound into the integrated energy identity gives

$$\|U(t)\|_2^2 + 2\nu \int_0^t \|\nabla U(s)\|_2^2 \, ds \leq 2 \int_0^t \|F(s)\|_2 N(s) \, ds = N(t)^2. \quad (17)$$

Since F is smooth and time compact, the right side is bounded independently of $t < 1$. Monotone convergence of the dissipation integral proves $D < \infty$.

The temporal support of F is a compact subset of $(0, \infty)$, so $F = 0$ on $[0, \tau]$ for some $\tau > 0$. Estimate (17) gives $U = 0$ there. Equation (15) then implies $\nabla P = 0$, and compact spatial support fixes this spatial constant to zero. Vanishing on an interval proves smoothness of both zero extensions. $\square$

Lemma 3.3 (Uniform localization of fractional norms). *Let B be a fixed coordinate ball whose closure lies inside a torus fundamental cube. If z is smooth and supported in B , let $z_{\mathbb{R}}$ be its zero extension in that cube to $\mathbb{R}^3$, and let $z_{\mathbb{T}^3}$ be its periodization. For $0 < s < 1$,*

$$\|z_{\mathbb{T}^3}\|_{H^s(\mathbb{T}^3)} \leq C_{s,B}(\|z_{\mathbb{R}}\|_2 + \|z_{\mathbb{R}}\|_{\dot{H}^s(\mathbb{R}^3)}). \quad (18)$$

The constant is uniform as the support shrinks inside B . At $s = 0$ the L^2 norms are equal; at $s = 1$ the corresponding equality holds for the L^2 gradient norms.

Proof. We first identify the difference expressions with the Fourier norms used in this article. Define

$$c_s = \int_{\mathbb{R}^3} \frac{|e^{ih_1} - 1|^2}{|h|^{3+2s}} dh.$$

It is positive and finite: near zero the numerator is at most $|h|^2$, giving a radial integral $\int_0^1 r^{1-2s} dr$; away from zero it is at most four, giving $\int_1^\infty r^{-1-2s} dr$. For a compact smooth Z on $\mathbb{R}^3$, change variables $y = x + h$, then use Plancherel and Tonelli on the nonnegative integrands to obtain

$$\begin{aligned} I_{\mathbb{R}}(Z) &:= \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \frac{|Z(x+h) - Z(x)|^2}{|h|^{3+2s}} dx dh \\ &= \int_{\mathbb{R}^3} |\widehat{Z}(\xi)|^2 \left(\int_{\mathbb{R}^3} \frac{|e^{i\xi \cdot h} - 1|^2}{|h|^{3+2s}} dh \right) d\xi = c_s \|Z\|_{\dot{H}^s(\mathbb{R}^3)}^2. \end{aligned}$$

For $\xi \neq 0$, rotation and dilation identify the inner integral with $c_s |\xi|^{2s}$; it is zero when $\xi = 0$.

Let Q be the chosen unit fundamental cube and define the nonnegative periodic kernel, off its lattice singularities, by

$$K_s(h) = \sum_{n \in \mathbb{Z}^3} |h + n|^{-3-2s}.$$

For a smooth periodic z , partitioning $\mathbb{R}^3$ into translates of Q and using periodicity gives

$$\begin{aligned} I_{\mathbb{T}^3}(z) &:= \int_Q \int_Q |z(x) - z(y)|^2 K_s(x - y) dx dy \\ &= \int_{\mathbb{R}^3} \frac{\|z(\cdot + h) - z\|_{L^2(Q)}^2}{|h|^{3+2s}} dh \\ &= \sum_{\ell \in \mathbb{Z}^3} |\widehat{z}(\ell)|^2 \int_{\mathbb{R}^3} \frac{|e^{2\pi i \ell \cdot h} - 1|^2}{|h|^{3+2s}} dh = c_s \|z - \widehat{z}(0)\|_{\dot{H}^s(\mathbb{T}^3)}^2. \end{aligned}$$

The second line uses ordinary periodic L^2 norms. Parseval proves the third line and Tonelli justifies interchanging the nonnegative sum and integral. Thus both identifications have the same explicitly specified constant c_s , with the frequency factor 2π retained on the torus. Also $(1 + r^2)^s \leq 1 + r^{2s}$ for $0 < s < 1$, so

$$\|z\|_{H^s(\mathbb{T}^3)}^2 \leq \|z\|_2^2 + c_s^{-1} I_{\mathbb{T}^3}(z).$$

It remains to compare the two difference integrals uniformly. Set $d = \text{dist}(\overline{B}, \partial Q) > 0$. For $x \in B$, $y \in Q$ and $n \neq 0$ we have $|x - y + n| \geq d$: the point $y - n$ lies outside Q up to its boundary. For $|n| > 2\sqrt{3}$ the same distance is at least $|n|/2$. Consequently

$$\sup_{x \in B, y \in Q} \sum_{n \neq 0} |x - y + n|^{-3-2s} \leq C_{s,d} < \infty,$$

and the identical estimate holds with x, y reversed. The $n = 0$ term of $I_{\mathbb{T}^3}(z_{\mathbb{T}^3})$ is at most $I_{\mathbb{R}}(z_{\mathbb{R}})$. For all the other terms use $|z(x) - z(y)|^2 \leq 2|z(x)|^2 + 2|z(y)|^2$; each nonzero term on

the right has its corresponding point in B . As $|Q| = 1$, their total is at most $4C_{s,d} \|z_{\mathbb{R}}\|_2^2$. This proves (18). Both d and all constants refer to B, Q, s , independently of any smaller support. At orders zero and one, integration over the single copy of the support proves the stated L^2 and gradient identities. $\square$

Choose a compact set K_* containing K and the spatial projection of $\text{supp } F$. Fix a point x_0 in a torus coordinate ball. For sufficiently small $\varepsilon > 0$, require

$$2\varepsilon^2 < T, \quad x_0 + \varepsilon K_* \subset B, \quad t_\varepsilon = T - \varepsilon^2,$$

and define, using the zero extensions from Lemma 3.2,

$$\begin{aligned} U_\varepsilon(x, t) &= \varepsilon^{-1} U\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right), \\ P_\varepsilon(x, t) &= \varepsilon^{-2} P\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right), \\ F_\varepsilon(x, t) &= \varepsilon^{-3} F\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right). \end{aligned} \tag{19}$$

The first two fields are used for $t < T$; the force is defined for all positive times. Place this single whole-space packet in B and then periodize. This operation does not rescale an already periodic field, which would replicate its support.

Proposition 3.4 (Packet scaling at fixed viscosity). *The fields in (19) solve the periodic momentum equation at viscosity ν , start from zero, and have unbounded speed at T . A spatially constant pressure adjustment enforces the mean-zero normalization. Their norms satisfy*

$$\|U_\varepsilon\|_{L^\infty(0,T;L^2)} = \varepsilon^{1/2} M, \quad \|\nabla U_\varepsilon\|_{L^2(0,T;L^2)} = \varepsilon^{1/2} D, \tag{20}$$

$$\|F_\varepsilon\|_{L^q(0,\infty;L^p)} = \varepsilon^{\alpha(p,q)} \|F\|_{L^q(0,\infty;L^p)}, \quad \alpha(p, q) = -3 + \frac{3}{p} + \frac{2}{q}, \tag{21}$$

for $1 \leq p, q \leq \infty$. For $0 \leq s \leq 1$,

$$\|F_\varepsilon\|_{L^1(0,\infty;H^s(\mathbb{T}^3))} \leq C_s (\varepsilon^{1/2} + \varepsilon^{1/2-s}). \tag{22}$$

In particular the force tends to zero in $L_t^1 H_x^s$ for every $s < 1/2$, including negative s .

Proof. Writing $y = (x - x_0)/\varepsilon$ and $\sigma = (t - t_\varepsilon)/\varepsilon^2$, every term of the momentum equation gains the common factor ε^{-3} . Incompressibility is preserved and no viscosity rescaling occurs. The temporal extension is smooth by initial vanishing. The speed identity $\|U_\varepsilon(t)\|_\infty = \varepsilon^{-1} \|U(\sigma)\|_\infty$ proves unboundedness as $t \uparrow T$.

At a fixed time the velocity has amplitude factor ε^{-1} and spatial volume factor ε^3 , giving the first identity in (20). Its gradient has amplitude factor ε^{-2} . Squaring and integrating in both space and time gives $\varepsilon^{-4} \varepsilon^3 \varepsilon^2 = \varepsilon$ times the original dissipation integral, proving the second identity.

For finite p, q , the force's spatial norm gains $\varepsilon^{-3+3/p}$ and its time norm gains $\varepsilon^{2/q}$. The positive-time domain includes the whole transformed support because $t_\varepsilon > 0$ and the original force vanishes at negative times. This proves (21); the essential-supremum cases follow by the same change of variables with zero reciprocal exponents.

Euclidean homogeneous Sobolev scaling gives, for $0 < s < 1$,

$$\left\| \varepsilon^{-3} F((\cdot - x_0)/\varepsilon, \sigma) \right\|_{\dot{H}^s(\mathbb{R}^3)} = \varepsilon^{-3/2-s} \|F(\cdot, \sigma)\|_{\dot{H}^s(\mathbb{R}^3)}.$$

One may verify the exponent either by Fourier transform or by the double integral in Lemma 3.3: its squared scaling factor is $\varepsilon^{-6} \varepsilon^6 \varepsilon^{-3-2s} = \varepsilon^{-3-2s}$. Time integration contributes ε^2 . The

inhomogeneous L^2 term has order $\varepsilon^{1/2}$. Lemma 3.3 therefore proves (22). At $s = 0$ use the mixed-norm identity, and at $s = 1$ use the gradient scaling. For $s < 0$, the Fourier definition gives $\|z\|_{H^s} \leq \|z\|_2$, so the $s = 0$ estimate proves convergence without any homogeneous negative-order claim. $\square$

The force peak grows as $\varepsilon^{-3} \|F\|_\infty$. Thus the smallness obtained here is smallness of an integrated norm. At the critical exponent, the Euclidean $L_t^1 \dot{H}_x^{1/2}$ norm is scale invariant; the independent regularity argument in Section 6 will turn that scaling boundary into a genuine non-density statement.

4 Exact insertion into a regular background

Fix a reference solution (v, π, g) smooth on $[0, T + \delta]$, where $\delta > 0$, $g \in \mathcal{F}$, and $v(0) = a \in \mathcal{X}$. Direct addition of U_ε would produce cross terms involving the singular packet and the nonzero reference. Those terms need not extend smoothly through T . We instead remove the reference in a small neighborhood of the packet support before adding the packet.

Lemma 4.1 (A local divergence-free cutoff). *Let v be smooth and divergence free in a spatial ball centered at x_0 . In that ball, with $y = x - x_0$, define*

$$A(x, t) = \int_0^1 r v(x_0 + ry, t) \times y \, dr. \quad (23)$$

Then $\nabla \times A = v$. Choose $\theta \in C_c^\infty(\mathbb{R}^3)$ equal to one on a neighborhood of K_ , and choose $\eta \in C_c^\infty((-2, 2))$ equal to one on $[-1, 1]$. Put*

$$\theta_\varepsilon(x) = \theta((x - x_0)/\varepsilon), \quad \eta_\varepsilon(t) = \eta((t - T)/\varepsilon^2), \quad w_\varepsilon = -\nabla \times (\eta_\varepsilon \theta_\varepsilon A). \quad (24)$$

For sufficiently small ε , w_ε is a smooth, divergence-free field supported inside the coordinate ball and in $(T - 2\varepsilon^2, T + 2\varepsilon^2)$. During the active presingular interval of U_ε ,

$$v + w_\varepsilon = 0 \quad \text{on an open neighborhood of } \text{supp } U_\varepsilon(t). \quad (25)$$

Proof. Use the vector identity

$$\nabla \times (V \times y) = V \nabla \cdot y - y \nabla \cdot V + (y \cdot \nabla)V - (V \cdot \nabla)y.$$

For $V(y) = v(x_0 + ry, t)$, incompressibility gives $\nabla_y \cdot V = 0$, $\nabla_y \cdot y = 3$, and $(V \cdot \nabla)y = V$. Consequently,

$$\begin{aligned} \nabla_y \times (r v(x_0 + ry, t) \times y) &= 2r v(x_0 + ry, t) + r^2 (y \cdot \nabla_x) v(x_0 + ry, t) \\ &= \frac{d}{dr} (r^2 v(x_0 + ry, t)). \end{aligned}$$

Integrating from $r = 0$ to $r = 1$ proves $\nabla \times A = v$.

Take ε small enough that the scaled support of θ is strictly inside the coordinate ball and $2\varepsilon^2 < \min(T, \delta)$. Since the spatial cutoff is zero near the boundary of the coordinate ball, extending (24) by zero gives a globally smooth periodic field. Its divergence vanishes because it is a curl. The temporal support follows directly from η_ε .

The packet is zero before $t_\varepsilon = T - \varepsilon^2$. For $t_\varepsilon \leq t < T$, the variable $(t - T)/\varepsilon^2$ belongs to $[-1, 0)$, on which $\eta = 1$. On a neighborhood of the packet support, $\theta_\varepsilon = 1$ and its derivatives vanish. There $w_\varepsilon = -\nabla \times A = -v$, proving (25). $\square$

Lemma 4.2 (Bounds for the background correction). *For w_ε in Lemma 4.1, define*

$$H_\varepsilon = \partial_t w_\varepsilon - \nu \Delta w_\varepsilon + (v \cdot \nabla) w_\varepsilon + (w_\varepsilon \cdot \nabla) v + (w_\varepsilon \cdot \nabla) w_\varepsilon. \quad (26)$$

The force H_ε is smooth across T and extends by zero to all times outside its cutoff support. Its spatial support has volume $O(\varepsilon^3)$, and its temporal support has length $O(\varepsilon^2)$. For every spatial multi-index β and integer $j \geq 0$,

$$|\partial_t^j \partial_x^\beta w_\varepsilon| \leq C_{\beta,j} \varepsilon^{-2j-|\beta|}, \quad |\partial_x^\beta H_\varepsilon| \leq C_\beta \varepsilon^{-2-|\beta|}. \quad (27)$$

In particular,

$$\|w_\varepsilon\|_{E_T} \leq C \varepsilon^{3/2}, \quad (28)$$

$$\|H_\varepsilon\|_{L^q(0,\infty;L^p)} \leq C_{p,q} \varepsilon^{-2+3/p+2/q} = C_{p,q} \varepsilon^{\alpha(p,q)+1}, \quad (29)$$

$$\|H_\varepsilon\|_{L^1(0,\infty;H^s)} \leq C_s (\varepsilon^{3/2} + \varepsilon^{3/2-s}) \quad (0 \leq s \leq 1). \quad (30)$$

All constants are independent of sufficiently small ε .

Proof. Introduce $z = (x - x_0)/\varepsilon$ and $\sigma = (t - T)/\varepsilon^2$. Then

$$A(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) = \varepsilon \mathcal{A}_\varepsilon(z, \sigma),$$

where

$$\mathcal{A}_\varepsilon(z, \sigma) = \int_0^1 r v(x_0 + \varepsilon r z, T + \varepsilon^2 \sigma) \times z \, dr.$$

On the fixed support of $\theta(z)\eta(\sigma)$, every fixed-order derivative of $\mathcal{A}_\varepsilon$ is bounded uniformly in ε . Differentiation of v in z or σ introduces nonnegative powers of ε , and the reference is smooth on a fixed neighborhood of the relevant spacetime set. Thus

$$w_\varepsilon(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) = W_\varepsilon(z, \sigma) := -\nabla_z \times (\eta(\sigma)\theta(z)\mathcal{A}_\varepsilon(z, \sigma))$$

is a uniformly smooth family supported in a fixed cylinder. This proves the first bound in (27).

To check the force bound, set $V_\varepsilon(z, \sigma) = v(x_0 + \varepsilon z, T + \varepsilon^2 \sigma)$. Formula (26) becomes

$$\begin{aligned} H_\varepsilon(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) &= \varepsilon^{-2} [\partial_\sigma W_\varepsilon - \nu \Delta_z W_\varepsilon + \varepsilon (V_\varepsilon \cdot \nabla_z) W_\varepsilon \\ &\quad + \varepsilon^2 (W_\varepsilon \cdot \nabla_x) v(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) + \varepsilon (W_\varepsilon \cdot \nabla_z) W_\varepsilon]. \end{aligned}$$

The expression in brackets and all its fixed spatial derivatives are uniformly bounded on the fixed cylinder. This proves the second bound in (27). It also makes explicit that H_ε involves only the smooth reference and fixed cutoffs, not derivatives of a singular field.

From the support volume and duration,

$$\|w_\varepsilon\|_{L_t^\infty L_x^2} \leq C \varepsilon^{3/2}, \quad \|\nabla w_\varepsilon\|_{L_t^2 L_x^2} \leq C \varepsilon^{-1} \varepsilon^{3/2} \varepsilon = C \varepsilon^{3/2}.$$

This proves (28). The amplitude bound $C \varepsilon^{-2}$ gives (29) by the same support calculation, including the essential-supremum endpoints.

For (30), the rescaled force profiles have uniformly bounded spatial H^s norms, supported on a fixed spatial set during a fixed interval of σ . An amplitude ε^{-2} gives the spatial homogeneous factor $\varepsilon^{-1/2-s}$; time integration adds ε^2 . Thus the homogeneous contribution is $O(\varepsilon^{3/2-s})$, while the L^2 contribution is $O(\varepsilon^{3/2})$. Lemma 3.3 transfers these estimates to the torus. Smooth zero extension in time is valid because η , with all its derivatives, vanishes near the fixed interval endpoints $\sigma = -2$ and $\sigma = 2$, where the reference remains smooth. $\square$

Theorem 4.3 (Exact local insertion). *Let (v, π, g) be a periodic solution smooth on $[0, T + \delta]$, with $\delta > 0$, $g \in \mathcal{F}$ and $v(0) = a \in \mathcal{X}$. Fix any nonempty coordinate ball. For all sufficiently small $\varepsilon > 0$, there are $g_\varepsilon \in \mathcal{F}$ and a solution u_ε with*

- (i) $T_{\max}^\nu(a, g_\varepsilon) = T$ and $\limsup_{t \uparrow T} \|u_\varepsilon(t)\|_\infty = \infty$;
- (ii) $u_\varepsilon = v$ for $0 \leq t \leq T - 2\varepsilon^2$;
- (iii) $u_\varepsilon - v$ divergence free and supported, for every $t < T$, in a ball of diameter $O(\varepsilon)$ inside the chosen ball;
- (iv) the simultaneous bounds

$$\|u_\varepsilon - v\|_{E_T} \leq (M + D)\varepsilon^{1/2} + C\varepsilon^{3/2}, \quad (31)$$

$$\|g_\varepsilon - g\|_{L_t^q L_x^p} \leq C_{p,q}(\varepsilon^{\alpha(p,q)} + \varepsilon^{\alpha(p,q)+1}), \quad (32)$$

$$\|g_\varepsilon - g\|_{L_t^1 H_x^s} \leq C_s(\varepsilon^{1/2-s} + \varepsilon^{3/2-s}) \quad (0 \leq s < 1/2). \quad (33)$$

Here force norms are taken over $(0, \infty)$, and $\alpha(p, q) = -3 + 3/p + 2/q$ for $1 \leq p, q \leq \infty$. For $s < 0$, the force difference also tends to zero in $L_t^1 H_x^s$.

Proof. Choose the scaled packet and the correction from Lemmas 4.1–4.2, with all supports inside the prescribed ball, and set

$$u_\varepsilon = v + w_\varepsilon + U_\varepsilon, \quad p_\varepsilon = \pi + P_\varepsilon, \quad g_\varepsilon = g + H_\varepsilon + F_\varepsilon. \quad (34)$$

Initially use the compact representative P_ε ; subtract its spatial mean to normalize p_ε .

Let $b_\varepsilon = v + w_\varepsilon$. Expanding the reference equation and (26) gives

$$\partial_t b_\varepsilon + (b_\varepsilon \cdot \nabla) b_\varepsilon - \nu \Delta b_\varepsilon + \nabla \pi = g + H_\varepsilon.$$

Adding the packet equation produces the equation for u_ε except for

$$(b_\varepsilon \cdot \nabla) U_\varepsilon + (U_\varepsilon \cdot \nabla) b_\varepsilon.$$

Both terms are identically zero. On a neighborhood of $\text{supp } U_\varepsilon$ this follows from $b_\varepsilon = 0$, including its derivatives, by (25). Outside that support the packet and its derivatives vanish; smoothness ensures the same assertion at the support boundary. When the packet is inactive it is zero everywhere. Thus (2) holds exactly for (34) on $[0, T]$.

Each summand of the velocity is divergence free. The correction is zero for $t \leq T - 2\varepsilon^2$, and the packet is zero before $T - \varepsilon^2$, so the initial datum and the stated earlier history are unchanged. On the active packet support, $u_\varepsilon = U_\varepsilon$. Hence

$$\|u_\varepsilon(t)\|_\infty \geq \|U_\varepsilon(t)\|_\infty$$

and the maximum norm is unbounded as $t \uparrow T$.

The velocity is smooth on every closed presingular interval. The force F_ε is globally smooth by its original definition, and H_ε is globally smooth by Lemma 4.2. Their time supports remain compact subsets of $(0, \infty)$ when ε is sufficiently small. Therefore $g_\varepsilon \in \mathcal{F}$, independently of the absence of a classical extension of u_ε at T . Uniqueness in Proposition 2.2 identifies the constructed velocity with the maximal solution for (a, g_ε) up to every $T' < T$. Its lifespan is at least T , and the unbounded maximum norm excludes extension beyond T . Thus the lifespan is exactly T .

Finally, the triangle inequality, Proposition 3.4 and Lemma 4.2 give (31)–(33). In (33), the lower-order inhomogeneous terms are absorbed because $0 < \varepsilon \leq 1$ and $s \geq 0$. For negative s , use $\|z\|_{H^s} \leq \|z\|_2$ and the $s = 0$ estimate. This proves every assertion. $\square$

Corollary 4.4 (Interior no-slip insertion). *Let $\Omega \subset \mathbb{R}^3$ be a bounded box or a bounded smooth domain. Suppose (v, π, g) is a smooth no-slip solution through $T + \delta$, and g is smooth on $\bar{\Omega} \times [0, \infty)$ with temporal support compact in $(0, \infty)$. Here smoothness on a closed spacetime slab means restriction of a C^∞ field from an open neighborhood of that slab; this convention also specifies smoothness at the edges and corners of a box. We assume v, π have this regularity on $\bar{\Omega} \times [0, T + \delta]$, and g on each finite closed slab. The equation and incompressibility hold in Ω , and $v|_{\partial\Omega} = 0$. Pressure may be normalized by zero spatial mean. Existence of this compatible reference is an assumption of the corollary. The construction in Theorem 4.3 applies inside any prescribed interior ball, preserving the initial velocity and no-slip boundary values. The energy estimates use $L^2(\Omega)$ and the force estimates use $L^1(0, \infty; H^s(\Omega))$, with the restriction norm (6). For every real s , the domain and zero-extended force-difference norms are comparable by (7), with a constant independent of ε ; the convergence assertion remains $s < 1/2$.*

Proof. The vector potential and all cutoff estimates are local. Choose the correction and packet supports in a smaller closed ball strictly inside Ω . The computation of the momentum equation in Theorem 4.3 is unchanged. The new velocity equals the reference in a fixed boundary collar for all presingular times, so the no-slip boundary values are preserved. The force correction is supported in the same interior region and has the same globally smooth time extension.

For $0 \leq s < 1/2$, the Euclidean scaling proof directly bounds the zero extensions of the force differences in $L^1(0, \infty; H^s(\mathbb{R}^3))$. For $s < 0$, use $\|E_0(g_\varepsilon - g)\|_{H^s(\mathbb{R}^3)} \leq \|E_0(g_\varepsilon - g)\|_{L^2(\mathbb{R}^3)}$ at each time. All these force-difference supports, including those after T , lie in one fixed compact interior ball K for every sufficiently small ε . Consequently (7) gives the asserted domain estimates and the scale-independent comparison after time integration. The energy estimates follow by integrating over that ball. Finally, classical uniqueness on a no-slip domain follows from the same difference-energy calculation as in Proposition 2.2; its boundary terms vanish. Thus the constructed solution is singular exactly at T . $\square$

Remark 4.5 (Concentration and force amplitudes). The force convergence in Theorem 4.3 is compatible with diverging pointwise amplitudes. Since the original force is nonzero,

$$\|g_\varepsilon - g\|_{L_{t,x}^\infty} \geq \varepsilon^{-3} \|F\|_\infty - C\varepsilon^{-2} \rightarrow \infty.$$

The first term is the exact packet peak; the second bounds the entire background-force correction. The original force must be nonzero, since (17) would otherwise force $U = 0$. Thus this lower bound is nonvacuous. Each g_ε is smooth, but this family has no uniform force-amplitude or derivative bounds.

5 Density, trajectory closure and data projections

Proposition 5.1 (Density on every fixed initial-velocity slice). *For $a \in \mathcal{X}$, $\nu > 0$, $T > 0$ and $s < 1/2$,*

$$\forall g \in \mathcal{F} \ \forall \rho > 0 \ \exists f \in \mathcal{F} : \quad \|f - g\|_{L_t^1 H_x^s} < \rho, \quad T_{\max}^\nu(a, f) \leq T.$$

Proof. Fix g and ρ . There are two exhaustive cases.

If $T_{\max}^\nu(a, g) \leq T$, choose $f = g$. It already belongs to $\mathcal{B}_{\nu,a,T}$ and the norm difference is zero.

If $T_{\max}^\nu(a, g) > T$, choose $\delta > 0$ so that $T + \delta < T_{\max}^\nu(a, g)$, and use its smooth solution as the reference in Theorem 4.3. For every sufficiently small ε , that theorem supplies $g_\varepsilon \in \mathcal{B}_{\nu,a,T}$ with lifespan exactly T . For $0 \leq s < 1/2$, (33) tends to zero. For $s < 0$, its $s = 0$ case and the embedding $L^2 \hookrightarrow H^s$ do so. Choose ε so that the force difference is less than ρ and set $f = g_\varepsilon$.

The initial datum a and the parameters ν, T are unchanged in both cases. This is precisely density in the relative force norm topology. $\square$

Corollary 5.2 (A sufficient mixed-norm region). *For every fixed $a \in \mathcal{X}$, the set $\mathcal{B}_{\nu,a,T}$ is dense in $\mathcal{F}$ with its relative $L^q(0, \infty; L^p)$ topology whenever*

$$1 \leq p, q \leq \infty, \quad \frac{3}{p} + \frac{2}{q} > 3. \quad (35)$$

Proof. Use the same two cases as in Proposition 5.1. In the regular-reference case, (32) tends to zero because $\alpha(p, q) > 0$ and hence $\alpha(p, q) + 1 > 0$. $\square$

This region includes $L_t^1 L_x^2$ and $L_t^2 L_x^{4/3}$. It is a sufficient condition, not a classification of all mixed Lebesgue topologies. A nonpositive value of the packet's scaling exponent alone proves neither density nor non-density.

Corollary 5.3 (Strong closure in energy and dissipation). *Fix $a \in \mathcal{X}$. Let $\mathcal{R}_{a,T}$ be the trajectories with initial velocity a and forces in $\mathcal{F}$ that are smooth through T . Let $\mathcal{S}_{a,T}$ be the trajectories with the same initial velocity and forces in $\mathcal{F}$ that are smooth on $[0, T)$, have finite E_T norm, and have unbounded maximum velocity at T . Then*

$$\mathcal{R}_{a,T} \subseteq \overline{\mathcal{S}_{a,T}}^{E_T}. \quad (36)$$

For each reference pair (v, g) and every fixed $s < 1/2$, the approximating pairs can be chosen with

$$(u_\varepsilon, g_\varepsilon) \longrightarrow (v, g) \quad \text{in } E_T \times L^1(0, \infty; H^s).$$

Proof. A reference smooth through T is smooth through $T + \delta$ for some $\delta > 0$, by local continuation; equivalently one may take this as the meaning of smoothness through the endpoint. Apply Theorem 4.3. Its energy estimate gives convergence in E_T , and its force estimates give simultaneous convergence. The reference has finite E_T norm and the packet and correction do also, so u_ε belongs to the specified ambient energy space.

To verify the claimed interpretation of the norm, note that

$$\|z\|_{L^2(0,T;L^2)} \leq T^{1/2} \|z\|_{L^\infty(0,T;L^2)}.$$

Thus E_T controls both terms of $L_t^2 H_x^1$, while the usual sum norm controls E_T directly. This proves the equivalence used in (36). No endpoint value or classical continuation of u_ε is implied by convergence in this norm. $\square$

Proposition 5.4 (Projection of extended singular data). *Define*

$$\mathfrak{B}_{\nu,T} = \{(a, f) \in \mathcal{X} \times \mathcal{F} : T_{\max}^\nu(a, f) \leq T\}.$$

If $s < 1/2$, this set is dense in the product of any topology on $\mathcal{X}$ and the relative $L_t^1 H_x^s$ topology on $\mathcal{F}$. Its projection onto $\mathcal{X}$ is all of $\mathcal{X}$. The family obtained while requiring $a = 0$ projects instead to the singleton $\{0\}$.

Proof. Take a point (a, g) and a basic product neighborhood $U \times V$ containing it. The initial datum a already belongs to U . Proposition 5.1 supplies a force $f \in V$ with $T_{\max}^\nu(a, f) \leq T$. Hence $(a, f) \in \mathfrak{B}_{\nu,T} \cap (U \times V)$, proving density. Applying the same proposition with any fixed g and any positive radius gives at least one singular force over each a , which proves the projection statement. The final assertion follows directly from fixing the initial velocity to zero. $\square$

The projection statement has the quantifier order $\forall a \exists f$, not $\exists f \forall a$. For a fixed prescribed force f_* , the set

$$\{a \in \mathcal{X} : T_{\max}^\nu(a, f_*) \leq T\}$$

is a different object and is not classified by these force-changing constructions. In a nontrivial normed initial-velocity space, $\{0\}$ is not dense: if $a \neq 0$, the ball of radius $\|a\|/2$ centered at a excludes zero.

There is also a distinction between the two terminal-time assertions. Theorem 4.3 gives singularity exactly at T around every reference regular through T . Proposition 5.1 uses breakdown by T , because a general reference may already break down earlier and is then left unchanged. It does not prove density of exactly-time- T singular forces around every earlier-breaking reference.

6 Critical forcing and a regular open neighborhood

The positive result does not decide what happens at $s = 1/2$. We now prove a regularity estimate that is independent of the packet construction. The critical Sobolev exponent is consistent with the classical Fujita–Kato framework [4]; the forcing, mean mode and continuation steps needed here are included explicitly.

Proposition 6.1 (Global regularity for small critical force). *There is $c > 0$, depending only on the unit torus and the stated norm conventions, such that*

$$g \in \mathcal{F}, \quad \rho := \|g\|_{L^1(0,\infty;H^{1/2})} < c\nu \quad (37)$$

imply $T_{\max}^\nu(0, g) = \infty$.

Proof. Let u be the maximal classical solution from Proposition 2.2. All estimates are first made on a compact subinterval of its lifespan.

Removal of the spatial mean. Define

$$\bar{g}(t) = \int_{\mathbb{T}^3} g(x, t) \, dx, \quad m(t) = \int_{\mathbb{T}^3} u(x, t) \, dx, \quad v = u - m, \quad h = g - \bar{g}.$$

Integrating the periodic equation eliminates the convective, Laplacian and pressure terms, giving $m' = \bar{g}$ and $m(0) = 0$. Thus

$$|m(t)| \leq \int_0^t |\bar{g}(s)| \, ds \leq \rho. \quad (38)$$

The mean-zero field v solves

$$\partial_t v + (v \cdot \nabla)v + (m \cdot \nabla)v - \nu \Delta v + \nabla p = h. \quad (39)$$

For any real Sobolev order, the spatially constant transport $m(t) \cdot \nabla$ commutes with the corresponding Fourier multiplier and is skew-adjoint in L^2 . It therefore contributes zero to both energy identities used below. This step is necessary because the force is not assumed to have zero spatial mean.

Critical energy estimate. Let $\Lambda = (-\Delta)^{1/2}$ on mean-zero fields, and set

$$y(t) = \left\| \Lambda^{1/2} v(t) \right\|_2, \quad z(t) = \left\| \Lambda^{3/2} v(t) \right\|_2, \quad b(t) = \|h(t)\|_{\dot{H}^{1/2}}.$$

Testing (39) against Λv gives

$$\frac{1}{2}(y^2)' + \nu z^2 = -\langle (v \cdot \nabla)v, \Lambda v \rangle + \langle h, \Lambda v \rangle.$$

The pressure term is zero because $\nabla \cdot \Lambda v = 0$. By Lemma 2.1 and Hölder's inequality,

$$|\langle (v \cdot \nabla)v, \Lambda v \rangle| \leq \|v\|_3 \|\nabla v\|_3 \|\Lambda v\|_3 \leq C_0 y z^2.$$

The force term is bounded by

$$|\langle h, \Lambda v \rangle| = |\langle \Lambda^{1/2} h, \Lambda^{1/2} v \rangle| \leq b(t)y(t).$$

Consequently,

$$\frac{1}{2}(y^2)' + (\nu - C_0 y)z^2 \leq by. \quad (40)$$

Removing the zero Fourier mode cannot increase the relevant norm, so

$$\int_0^\infty b(t) dt \leq \rho. \quad (41)$$

Closing the bootstrap, including the zero norm. On any interval where $y \leq \nu/(2C_0)$, (40) implies $\frac{1}{2}(y^2)' \leq by$. For $\zeta > 0$,

$$\frac{d}{dt}(y^2 + \zeta^2)^{1/2} \leq b \frac{y}{(y^2 + \zeta^2)^{1/2}} \leq b.$$

Since $y(0) = 0$, integration and $\zeta \downarrow 0$ yield

$$y(t) \leq \int_0^t b(s) ds \leq \rho \quad (42)$$

throughout such an interval. Choose $c < 1/(4C_0)$. If a first time with $y = \nu/(2C_0)$ existed, continuity would permit the preceding calculation up to that time, where it would give the strictly smaller bound $y \leq \rho < \nu/(4C_0)$. This contradiction closes the bootstrap on the whole classical lifespan.

In particular the critical dissipation also obeys

$$\frac{1}{2}y(t)^2 + \frac{\nu}{2} \int_0^t z(s)^2 ds \leq \int_0^t b(s)y(s) ds \leq \rho^2.$$

We next obtain a stronger integrability bound that directly implies classical continuation.

The H^1 estimate. Take the inner product of (39) with $-\Delta v$. The pressure and mean-transport terms vanish. The nonlinear term satisfies

$$\begin{aligned} |\langle (v \cdot \nabla)v, \Delta v \rangle| &\leq \|v\|_3 \|\nabla v\|_6 \|\Delta v\|_2 \\ &\leq C_1 y \|\Delta v\|_2^2. \end{aligned}$$

Here $\|\nabla v\|_6 \leq C \|\Delta v\|_2$ follows from Lemma A.1, applied to each mean-zero derivative: $\|\nabla(\partial_j v)\|_2 \leq \|\Delta v\|_2$ by its Fourier series. Choose also $c < 1/(4C_1)$. By (42), the convection term is at most $(\nu/4) \|\Delta v\|_2^2$. Young's inequality bounds the force term by $(\nu/4) \|\Delta v\|_2^2 + C\nu^{-1} \|h\|_2^2$. After multiplying the resulting inequality by two and weakening its positive coefficient if necessary,

$$(\|\nabla v\|_2^2)' + \nu \|\Delta v\|_2^2 \leq C\nu^{-1} \|h\|_2^2. \quad (43)$$

Since $v(0) = 0$, integration gives a uniform bound on $\|\nabla v(t)\|_2^2$ and on $\int_0^t \|\Delta v(s)\|_2^2 ds$. The integral of $\|h\|_2^2$ over all positive times is finite because each fixed force is smooth and time compact. It need not be small.

Continuation. Suppose the maximal lifespan were a finite number S . For mean-zero v , Fourier series give $\|v\|_{H^2} \leq C \|\Delta v\|_2$. Together with (43) and (38), the orthogonality of the constant and mean-zero Fourier modes gives

$$\begin{aligned} \int_0^S \|u(t)\|_{H^2}^2 dt &= \int_0^S (|m(t)|^2 + \|v(t)\|_{H^2}^2) dt \\ &\leq S\rho^2 + C\nu^{-2} \int_0^\infty \|h(t)\|_2^2 dt < \infty. \end{aligned}$$

Proposition 2.2 then extends u beyond S , a contradiction. Thus the maximal lifespan is infinite. The choice of c uses only the constants in the periodic Sobolev estimates, not the amplitude, derivatives or support size of a particular force. $\square$

Corollary 6.2 (Non-density at and above the critical exponent). *For every $s \geq 1/2$ and $T > 0$, $\mathcal{B}_{\nu,T}^0$ is not dense in $\mathcal{F}$ with the relative $L_t^1 H_x^s$ topology.*

Proof. At $s = 1/2$, the set

$$\{g \in \mathcal{F} : \|g\|_{L_t^1 H_x^{1/2}} < c\nu\}$$

is a nonempty relative open ball, containing zero, and is disjoint from $\mathcal{B}_{\nu,T}^0$ by Proposition 6.1. For $s \geq 1/2$, the Fourier weights imply $\|g(t)\|_{H^{1/2}} \leq \|g(t)\|_{H^s}$. The ball $\|g\|_{L_t^1 H_x^s} < c\nu$ is therefore also disjoint from the singular-force set. A dense subset cannot miss a nonempty open set. $\square$

Proof of Theorem 1.1. Proposition 5.1 gives density for every fixed smooth initial velocity when $s < 1/2$. Corollary 6.2 gives non-density on the zero-data slice when $s \geq 1/2$. These are the two assertions of the theorem. $\square$

The non-density statement is proved for the zero-data slice. It does not classify every nonzero-data slice at or above the critical exponent. In contrast, the positive insertion preserves any fixed smooth initial velocity.

7 Additional singular families

Proposition 7.1 (Infinite-dimensional variations away from the endpoints). *Fix the packet (U, P, F) of Theorem 3.1, and fix a spacetime cylinder $Q = B_0 \times (\tau_0, \tau_1)$ with $0 < \tau_0 < \tau_1 < 1$ and $B_0 \Subset \mathbb{R}^3$ an open ball. For any $b \in C_c^\infty(Q; \mathbb{R}^3)$ with $\nabla \cdot b = 0$, define*

$$\begin{aligned}\tilde{U} &= U + b, & \tilde{P} &= P, \\ \tilde{F} &= F + \partial_t b - \nu \Delta b + (U \cdot \nabla)b + (b \cdot \nabla)U + (b \cdot \nabla)b.\end{aligned}\tag{44}$$

These fields give an infinite-dimensional affine family of distinct velocities with zero initial data, finite energy and dissipation, and the same late singular behavior as U . For each fixed nonzero b , the family obtained by replacing b with λb is non-isolated as $\lambda \rightarrow 0$ in every fixed smooth seminorm of the velocity and force differences on their compact support.

Proof. Expanding the momentum equation for $U + b$ gives exactly the force in (44); its divergence remains zero. Every correction term is supported in $\text{supp } b$, a compact subset of Q . The original packet is smooth with bounded derivatives of every fixed order on a neighborhood of that set. Thus the force correction extends smoothly by zero to the whole positive time axis, including time one, despite the singularity of U there.

Since $b = 0$ near time zero and near time one, $\tilde{U}(0) = 0$ and $\tilde{U} = U$ on a fixed late interval. The same unbounded-speed limit holds. The energy and dissipation norms of b are finite by compact smooth support, so the triangle inequality gives the corresponding bounds for $\tilde{U}$.

To verify infinite dimensionality explicitly, choose countably many disjoint small balls inside B_0 . In each ball choose a smooth compactly supported vector potential with nonzero curl, and multiply that curl by a fixed nonzero time bump in (τ_0, τ_1) . These divergence-free fields are linearly independent because their spatial supports are disjoint. Their finite linear combinations belong to the allowed class. The map $b \mapsto U + b$ is injective and affine, so its image is infinite dimensional.

For the non-isolation statement, the velocity difference is λb , while the force difference has the form

$$\lambda L_U b + \lambda^2 (b \cdot \nabla)b, \quad L_U b = \partial_t b - \nu \Delta b + (U \cdot \nabla)b + (b \cdot \nabla)U.$$

On the fixed compact support, all coefficients and their derivatives are bounded. Hence for each integer $m \geq 0$ the C^m norm of the force difference is at most $C_m|\lambda| + C'_m\lambda^2$, and the velocity difference is at most $C''_m|\lambda|$. Both tend to zero. The forces vary in this family; no open singularity basin for one fixed force follows. $\square$

Proposition 7.2 (Finitely many prescribed singular regions). *Fix finitely many disjoint interior balls $B_1, \dots, B_N$ in the torus or in a bounded three-dimensional domain. At fixed $\nu > 0$ and $T > 0$, there is a smooth forced solution starting from rest, smooth on $[0, T)$, with finite energy and dissipation, such that*

$$\limsup_{t \uparrow T} \|u(t)\|_{L^\infty(B_j)} = \infty \quad (1 \leq j \leq N).$$

In a bounded domain its boundary condition is homogeneous no-slip.

Proof. Choose $x_j \in B_j$ and scales $\varepsilon_j > 0$ sufficiently small that the complete scaled compact set $x_j + \varepsilon_j K_*$ lies strictly inside B_j and $\varepsilon_j^2 < T$. Apply (19) to each packet with start time $T - \varepsilon_j^2$, and denote the resulting fields by U_j, P_j, F_j . Set

$$u = \sum_{j=1}^N U_j, \quad p = \sum_{j=1}^N P_j, \quad f = \sum_{j=1}^N F_j.$$

For $i \neq j$, the supports of U_i and U_j are separated, so $(U_i \cdot \nabla)U_j = 0$. The sum therefore satisfies the momentum equation exactly. All force supports are compact in positive time and all forces are smooth. Initial vanishing gives $u(0) = 0$.

On B_j the velocity equals U_j , proving the stated separate limsup for each ball. Disjoint supports also give

$$\sup_{t < T} \|u(t)\|_2^2 \leq M^2 \sum_{j=1}^N \varepsilon_j, \quad \int_0^T \|\nabla u(t)\|_2^2 dt = D^2 \sum_{j=1}^N \varepsilon_j.$$

These sums are finite. In a bounded domain, all fields vanish in a boundary collar; hence $u = 0$ on the boundary. On the torus a spatially constant pressure adjustment gives the chosen gauge. $\square$

The proposition concerns a finite number of packets with a common terminal time. It asserts neither an infinite superposition nor a classical continuation after the first singular endpoint. The times at which individual packet maxima are large need not coincide; the displayed conclusions are separate limsup statements as $t \uparrow T$.

Proposition 7.3 (Conservative forcing from rest). *On the torus, let $f = -\nabla\phi$ for a globally defined periodic scalar ϕ . On a bounded domain, let $f = -\nabla\phi$ and impose homogeneous no-slip velocity. In either case, any smooth solution starting from rest is identically zero on its classical lifespan.*

Proof. Incompressibility and periodicity or the zero normal boundary trace give

$$\int f \cdot u = - \int \nabla\phi \cdot u = 0.$$

The classical energy identity, integrated from zero, becomes

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = 0.$$

Both terms are nonnegative, so $u(t) = 0$ at every presingular time. The periodic-potential condition excludes affine potentials, whose gradients need not be removable by the allowed periodic pressure gauge. $\square$

Every nontrivial from-rest construction in this article therefore requires a nonconservative force. Smoothness of the zero initial velocity does not imply that the resulting force-velocity pair is regular through the terminal time.

8 Discussion

The main theorem gives a complete dense/not-dense classification in a particular scale of force topologies on the zero-initial-velocity slice. Below $s = 1/2$, the force norm permits enough concentration for a singular packet to be inserted near any regular reference. At and above the critical exponent, an independent estimate provides an open set of globally regular smooth forces. The critical conclusion is an analytic obstruction, not an inference drawn solely from scaling.

The trajectory result is equally specific. Singular solutions can approach a regular reference strongly in both the kinetic-energy and integrated-dissipation terms of E_T . At each presingular time they are smooth and belong to every finite-order spatial Sobolev space. These facts do not bound their maximum velocity uniformly as the terminal time is approached.

The extended-data projection reaches every smooth initial velocity because the force is free to depend on that velocity. This does not determine the density, dimension or stability of singular initial states under a single prescribed force. The literal from-rest family has only the singleton initial datum. Maintaining these quantifiers is essential to any interpretation of the distribution result.

No probability law on forces or perturbations has been specified. Density does not establish typicality, an open basin of singularity, or robustness under arbitrary errors. Likewise, density in an infinite-dimensional input space does not imply density within a fixed spectral truncation or actuator family. The approximating forces have shorter scales and increasing peaks. A numerical conclusion must therefore identify the retained observations and the error norm.

The results do not select a continuation after the singular time. No CFD experiment is claimed.

Declaration of generative AI use

GPT Astra (OpenAI) was used to assist with literature and Lean-source inspection, development and checking of mathematical arguments, adversarial review, drafting, and LaTeX preparation. The use of generative AI does not constitute formal verification. Responsibility for the final content remains with the authors.

A A direct proof of the critical embeddings

This appendix proves the two critical embeddings used in the nonlinear estimates. Only elementary integration, Fourier inversion and Plancherel, Hölder's inequality, and approximation in L^2 are used. In particular, no Hardy–Littlewood–Sobolev or Sobolev embedding theorem is assumed. Throughout this appendix $\Lambda = (-\Delta)^{1/2}$, with Fourier symbol $|\xi|$ on $\mathbb{R}^3$ and $2\pi|k|$ on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$.

Lemma A.1 (Critical embeddings on the two domains). *For $a \in \{1/2, 1\}$ and $p_a = 6/(3 - 2a)$, there are finite constants, depending only on a and on the fixed unit torus when relevant, such that*

$$\|v\|_{L^{p_a}(\mathbb{R}^3)} \leq C_a \|\Lambda^a v\|_{L^2(\mathbb{R}^3)}, \quad \|v\|_{L^{p_a}(\mathbb{T}^3)} \leq C'_a \|\Lambda^a v\|_{L^2(\mathbb{T}^3)}. \quad (45)$$

The first inequality holds initially for Schwartz functions and extends to the homogeneous completion specified below. The second holds for mean-zero periodic distributions with finite

displayed right-hand side; the mean-zero condition is essential. Componentwise application gives the same statements for vector and tensor fields, with constants allowed to depend on their fixed number of components.

Consequently, on either domain and whenever the indicated homogeneous norms are finite (with mean-zero representatives on the torus),

$$\begin{aligned} \|v\|_3 &\leq C \|v\|_{\dot{H}^{1/2}}, \\ \|\nabla v\|_3 + \|\Lambda v\|_3 &\leq C \|v\|_{\dot{H}^{3/2}}, \\ \|\nabla v\|_6 &\leq C \|\Delta v\|_2. \end{aligned} \tag{46}$$

Here the derivative statements on $\mathbb{R}^3$ refer to distributions whose derivatives have the homogeneous realizations described in the proof; in particular they apply to all the smooth H^∞ fields used in the article.

Proof. 1. A maximal-function estimate, including its proof. For a locally integrable g on $\mathbb{R}^3$ let

$$Mg(x) = \sup_{r>0} \frac{1}{|B(x,r)|} \int_{B(x,r)} |g(y)| \, dy.$$

For $g \in L^1$ and $\lambda > 0$, the set $E_\lambda = \{Mg > \lambda\}$ is open: for each fixed radius the ball integral is continuous in its center, by translation continuity in L^1 on bounded sets. Every compact $K \subset E_\lambda$ is covered by finitely many balls whose averages exceed λ . Select the largest remaining ball and discard all balls meeting it, then repeat. The selected balls B_j are disjoint, and each discarded ball lies in the concentric triple of the ball that discarded it. Therefore

$$|K| \leq \sum_j |3B_j| = 27 \sum_j |B_j| \leq \frac{27}{\lambda} \|g\|_1.$$

Exhaustion by compact subsets proves $|E_\lambda| \leq 27\lambda^{-1} \|g\|_1$.

For $g \in L^2$, split $g = g\mathbf{1}_{\{|g| \leq \lambda/2\}} + g\mathbf{1}_{\{|g| > \lambda/2\}}$. The first part has maximal function at most $\lambda/2$, and the second is in L^1 . The preceding estimate gives

$$|\{Mg > \lambda\}| \leq \frac{54}{\lambda} \int_{\{|g| > \lambda/2\}} |g(x)| \, dx.$$

The layer-cake identity and Tonelli's theorem now yield

$$\|Mg\|_2^2 = 2 \int_0^\infty \lambda |\{Mg > \lambda\}| \, d\lambda \leq 108 \int_{\mathbb{R}^3} |g(x)| \int_0^{2|g(x)|} d\lambda \, dx = 216 \|g\|_2^2. \tag{47}$$

This also proves that Mg is finite almost everywhere; no prior integrability of Mg was needed.

2. *The potential kernel from the Fourier multiplier.* For $0 < a < 3$, define I_a by the multiplier $|\xi|^{-a}$ on Schwartz functions. The Gamma integral gives, for $\xi \neq 0$,

$$|\xi|^{-a} = \frac{1}{\Gamma(a/2)} \int_0^\infty t^{a/2-1} e^{-t|\xi|^2} \, dt, \quad \Gamma(b) = \int_0^\infty s^{b-1} e^{-s} \, ds.$$

Direct Gaussian Fourier integration shows that the heat multiplier $e^{-t|\xi|^2}$ acts by convolution with $h_t(x) = (4\pi t)^{-3/2} e^{-|x|^2/(4t)}$ in the unitary Fourier convention of this article. Substituting $s = |x|^2/(4t)$ gives

$$\begin{aligned} K_a(x) &= \frac{1}{\Gamma(a/2)} \int_0^\infty t^{a/2-1} h_t(x) \, dt \\ &= \frac{\Gamma((3-a)/2)}{2^a \pi^{3/2} \Gamma(a/2)} |x|^{a-3} =: c_a |x|^{a-3}, \quad x \neq 0. \end{aligned}$$

Thus $I_a g = K_a * g$. To justify the identity without a formal interchange, for Schwartz g the small- t integrand is bounded by $t^{a/2-1} \|g\|_\infty$, and the large- t integrand is bounded by $C t^{(a-3)/2-1} \|g\|_1$. Both are integrable in their respective ranges. The same bounds after pairing with a Schwartz test function justify the heat representation in distributions. On the Fourier side, $|\xi|^{-a}$ is locally integrable for $a < 3$ and the other factors decay rapidly, so Fubini gives exactly the displayed multiplier. On the spatial side the locally integrable kernel and the decay of g justify the convolution and its heat-integral representation.

3. *The potential estimate.* Take $0 < a < 3/2$, $g \in L^2$ and $R > 0$. Splitting the potential at radius R , and dividing the inner ball into the annuli with outer radii $2^{-j}R$, gives

$$\begin{aligned} \int_{|y| < R} |y|^{a-3} |g(x-y)| \, dy &\leq C_a \sum_{j=0}^{\infty} (2^{-j}R)^a M g(x) \leq C_a R^a M g(x), \\ \int_{|y| \geq R} |y|^{a-3} |g(x-y)| \, dy &\leq \|g\|_2 \left(4\pi \int_R^\infty r^{2a-4} \, dr \right)^{1/2} \leq C_a \|g\|_2 R^{a-3/2}. \end{aligned}$$

The first inequality uses the average over each outer ball and the maximum of the kernel on that annulus. The second integral converges precisely because $a < 3/2$. At points where $0 < M g(x) < \infty$ choose $R = (\|g\|_2 / M g(x))^{2/3}$. The zero function is harmless; if $M g(x) = 0$ at some point, the integrals over every ball centered there show that $g = 0$ almost everywhere. We have proved almost everywhere

$$|I_a g(x)| \leq C_a \|g\|_2^{2a/3} M g(x)^{1-2a/3}.$$

This also proves absolute convergence of the kernel integral almost everywhere for $g \in L^2$. With $p_a = 6/(3-2a)$, the identity $p_a(1-2a/3) = 2$ and (47) imply

$$\|I_a g\|_{p_a} \leq C_a \|g\|_2^{2a/3} \|M g\|_2^{1-2a/3} \leq C_a \|g\|_2. \quad (48)$$

4. *The Euclidean function class and passage to the limit.* First let v have Fourier transform in $C_c^\infty(\mathbb{R}^3 \setminus \{0\})$. Then $g = \Lambda^a v$ is Schwartz, the multiplier identity in step 2 gives $v = I_a g$, and (48) proves the Euclidean inequality. For clarity, the homogeneous space here is the completion of this class in $\|\Lambda^a v\|_2$, for $0 < a < 3/2$. It has a unique realization as distributions, and the estimate gives a unique L^{p_a} representative. Indeed, if $G \in L^2$ is the limit of $|\xi|^a \hat{v}_n$, its distributional representative has Fourier transform $|\xi|^{-a} G$: for every Schwartz test function ϕ ,

$$\int_{\mathbb{R}^3} |\xi|^{-a} |G(\xi) \hat{\phi}(\xi)| \, d\xi \leq \|G\|_2 \left(\int_{\mathbb{R}^3} |\xi|^{-2a} |\hat{\phi}(\xi)|^2 \, d\xi \right)^{1/2} < \infty.$$

The last integral is finite at zero because $2a < 3$, and finite at infinity by rapid decay. This estimate also proves distributional convergence. Meanwhile the embedding makes v_n Cauchy in L^{p_a} , and convergence in that space implies distributional convergence by Hölder's inequality; the two limits coincide. Every $G \in L^2$ is approximable by smooth functions supported in compact annuli: first truncate in frequency, then mollify within a slightly larger annulus. Hence this realization covers every such G , without an ambiguity by additive polynomials. In particular any Schwartz v , and any H^∞ field with the stated homogeneous norm, is covered by approximating $G = |\xi|^a \hat{v}$ in this way.

5. *Transfer to the unit torus.* Choose $\eta \in C_c^\infty(\mathbb{R}^3)$ equal to one on the cube $[-1/2, 1/2]^3$. For a trigonometric polynomial v , extend it periodically and set $V = \eta v$. Write $\langle \xi \rangle = (1 + |\xi|^2)^{1/2}$. Fourier transformation gives

$$\hat{V}(\xi) = \sum_{k \in \mathbb{Z}^3} \hat{v}(k) \hat{\eta}(\xi - 2\pi k).$$

Integration by parts in the compactly supported smooth function η gives arbitrarily fast decay of $\hat{\eta}$. Together with $\langle \xi \rangle^a \leq C_a \langle 2\pi k \rangle^a \langle \xi - 2\pi k \rangle^a$, this yields

$$\langle \xi \rangle^a |\hat{V}(\xi)| \leq C_{a,\eta} \sum_k A_k Q(\xi - 2\pi k), \quad A_k = \langle 2\pi k \rangle^a |\hat{v}(k)|, \quad Q(z) = (1 + |z|^2)^{-2}.$$

Here $Q \in L^1(\mathbb{R}^3)$, and $\sup_\xi \sum_k Q(\xi - 2\pi k) < \infty$: reduce ξ to a fixed lattice cell and group the lattice points into shells, whose contributions are bounded by $C \sum_{n \geq 0} (n+1)^2 (1+n^2)^{-2} < \infty$. Weighted Cauchy–Schwarz, followed by integration, therefore proves

$$\begin{aligned} \|V\|_{H^a(\mathbb{R}^3)}^2 &\leq C_{a,\eta} \int_{\mathbb{R}^3} \left(\sum_k A_k^2 Q(\xi - 2\pi k) \right) \left(\sum_k Q(\xi - 2\pi k) \right) d\xi \\ &\leq C_{a,\eta} \sum_k A_k^2 = C_{a,\eta} \|v\|_{H^a(\mathbb{T}^3)}^2. \end{aligned}$$

Apply the Euclidean estimate to V and restrict to the unit cell. If $\hat{v}(0) = 0$, the elementary spectral-gap estimate

$$(1 + 4\pi^2 |k|^2)^a \leq (1 + (2\pi)^{-2})^a (2\pi |k|)^{2a}, \quad k \neq 0,$$

then gives

$$\|v\|_{L^{pa}(\mathbb{T}^3)} \leq \|V\|_{L^{pa}(\mathbb{R}^3)} \leq C_a \|V\|_{\dot{H}^a(\mathbb{R}^3)} \leq C'_a \|v\|_{\dot{H}^a(\mathbb{T}^3)}.$$

Fourier truncations converge in $\dot{H}^a$ for every mean-zero periodic distribution with finite norm, hence also in L^{pa} by this estimate. Their distributional limits identify the two representatives. This proves the periodic statement, including its extension beyond polynomials.

Finally apply the $a = 1/2$ estimate to each $\partial_j v$ and to Λv , and the $a = 1$ estimate to each $\partial_j v$. Plancherel and $\sum_j |\xi_j|^2 = |\xi|^2$ (or its Fourier-series version) give (46). For smooth fields the resulting nonlinear bound is, explicitly,

$$| \langle (v \cdot \nabla) v, \Lambda v \rangle | \leq \|v\|_3 \|\nabla v\|_3 \|\Lambda v\|_3 \leq C \|v\|_{\dot{H}^{1/2}} \|v\|_{\dot{H}^{3/2}}^2.$$

Every constant above is independent of the field and of its frequency support. $\square$

References

- [1] OpenAI. *Finite Time Blowup for Navier–Stokes*. Manuscript, 166 pp., accessed 9 September 2026. Official manuscript.
- [2] OpenAI. *NavierStokesAndEuler*. Lean source repository, revision 8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538, accessed 9 September 2026. Source revision. Compact witness: `selected_compact_candidate`; its support and zero-data conditions: `R3CompactCandidate.Properties`.
- [3] M. Hofmanová, R. Zhu and X. Zhu. *Non-uniqueness of Leray–Hopf solutions for stochastic forced Navier–Stokes equations*. arXiv:2309.03668v2, 10 September 2024. arXiv:2309.03668.
- [4] H. Fujita and T. Kato. On the Navier–Stokes initial value problem. I. *Archive for Rational Mechanics and Analysis* **16** (1964), 269–315. doi:10.1007/BF00276188.
- [5] T. Tao. Localisation and compactness properties of the Navier–Stokes global regularity problem. *Analysis & PDE* **6**(1) (2013), 25–107. doi:10.2140/apde.2013.6.25.
- [6] T. Tao. *254A, Notes 1: Local well-posedness of the Navier–Stokes equations*. Author’s course notes, 16 September 2018. Original text.